TEZPUR UNIVERSITY

Spring Semester End Examination, 2023

Course Code: CO206 **Course Title:** Design and Analysis of Algorithms

Full Marks: 60 **Time:** 3 hours

You may answer all the questions but maximum you can get is 60.

1. Following is a variation of Knapsack problem.

   Input: n items of non-negative weights $w_1, w_2, \cdots, w_n$, profits $p_1, p_2, \cdots, p_n$, two positive integers C and S.

   Output (yes/no:) Does there exist a subset of items with total weight at most C that correspond to a profit of S?

   Does the above problem belong to the class NP? [7]

2. Answer True / False and provide a counterexample or a short justification. Correct answers with incorrect justification will get only half of the marks. [5×5]

   a) Any n node tree ($n > 2$) has atleast 2 leaf nodes.

   b) If $T(n) = 3T(\lceil \frac{n-3}{5} \rceil) + O(n)$, then $T(n) \in \Theta(n^2)$.

   c) The largest weighted edge of a graph G will not be in the minimum spanning tree of G.

   d) Dijkstra's algorithm can find the longest path in a graph (with positive weights) from a source vertex if we multiply every edge weight with -1.

   e) An optimal solution to a 0/1 knapsack problem will always contain the object with the greatest benefit/weight ratio.

3. Assume there are n men with heights $m_1, \ldots, m_n$ and there are n women with heights $w_1, \ldots, w_n$. The task is to pair each man with a woman in such a way that the total height difference between the pairs is minimized. Each man or women is matched with exactly one partner. That means, the task is to minimize $\sum_{i=1}^{n} |m_i - w_j|$ where (i, j) are partners.

Consider the following two greedy algorithms:

(a) Find the man and woman with the minimum height difference. Pair them up. Repeat until every man and woman is matched.

(b) Sort the men and women by height. Match the shortest man with the shortest woman, and repeat.

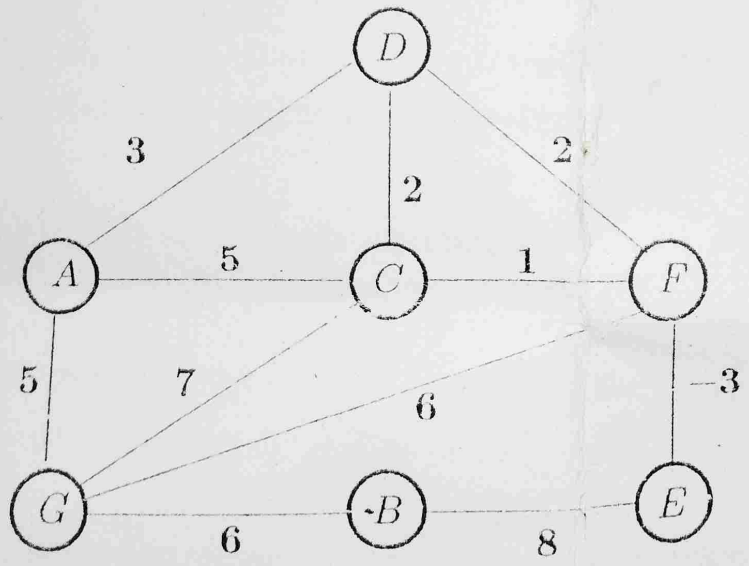
Which one of the algorithms above is correct? Disprove the incorrect one with a counterexample. For the correct algorithm, provide a proof of its correctness [2+5+8]

1. Suppose we have a bag of capacity 10 kg, which we want to pack with items of values $\{10, 40, 30, 50\}$ and weights 5, 4, 6 and 3 respectively. Our goal is to pack the bag with as many items as possible so that the total weight of the items does not exceed the capacity of the bag and the total value is maximized. Solve the problem using dynamic programming **(Show the table only)**. Will your algorithm work if:

- one of the item has negative weight?
- one of the item has negative weight and negative profit? [5+5+5]

a) What is the cost of a minimum spanning tree of the graph below? [3]

b) How many minimum spanning trees does this graph have? List them all. [5]

c) Run Prim's algorithm on this graph. For each of the iteration, show: The SCORE of each vertex, the HEAP, the vertex extracted from the HEAP. [8]

d) Which data structure will you use to store the MST? Explain. [4]